

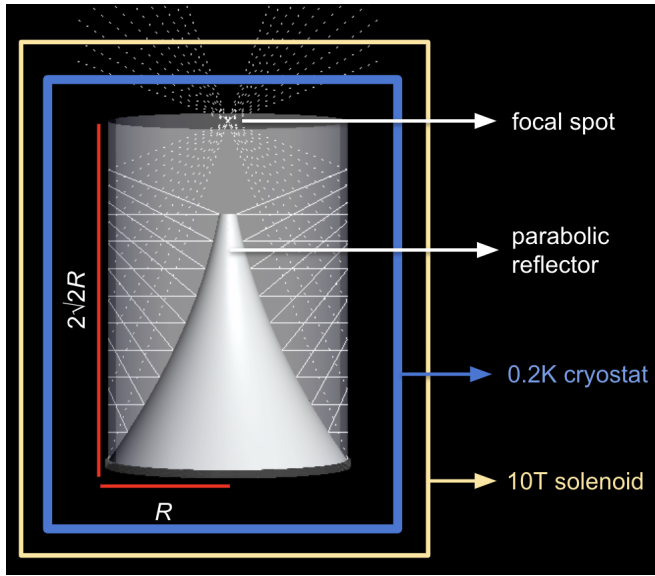


Figure 2: BREAD setup. For pilot experiments, $R = 201\text{mm}$.

post-inflationary models predict that the axion exists at higher frequencies above 10^{-3} GHz [5]. However, the resonant cavity approach becomes infeasible and impractical at these higher masses.

Instead, the Broadband Reflector Experiment for Axion Detection (BREAD) directly probes this higher frequency range using a “broadband” approach covering the range $[10^{-3}, 1]\text{eV}$, $[0.24, 240]\text{THz}$, and an innovative coaxial parabolic “reflector” to focus signal photons onto a detector [6]. A metal cylinder is placed inside of a solenoid producing a $\mathcal{O}(10\text{ T})$ magnetic field. The presence of an axion induces an oscillating electric field along the surface of the cylinder, which creates a compensating electromagnetic wave that propagates directly orthogonal to the tangent plane of the cylinder with the same frequency as the axion. Inside the metal cylinder, a metal parabolic reflector revolved around the axis perpendicular to its axis of symmetry reflects this emitted electromagnetic wave from any point along the height of the cylinder to a single focal spot at the top of the cylinder. BREAD will also be placed inside of a cryostat cooled to 0.2K to reduce thermal noise.

While photoconversion occurs universally across all frequencies, the optimal photosensor technology changes depending on the frequency of the photon. Currently, two experimental versions of BREAD with different photosensor technologies are being developed to look for axions in the low and high mass limits of the $[10^{-3}, 1]\text{eV}$ range: GigaBREAD and InfraBREAD, respectively. GigaBREAD will search for radio frequency axions at $\mathcal{O}(10\text{ GHz})$. In this low-mass limit where $\lambda_{dB} \gg R$, the de Broglie wavelength of the axion is much greater than the radius of the cylinder and a coaxial horn antenna is used as a detector. InfraBREAD will search for infrared frequency axions at $\mathcal{O}(200\text{ THz})$. In this high-mass limit

where $\lambda_{dB} \ll R$, a single-photon counting detector called an Superconducting Nanowire Single Photon Detector (SNSPD) is used. SNSPDs fabricated at JPL and MIT are 1 mm^2 detectors containing a grid of eight sub pixels each with layers of sub-micrometer width superconducting wires. When a photon above a certain energy threshold collides with one of these nanowires, superconductivity is momentarily abruptly and a signal can be observed [7].

A. Purpose of Study

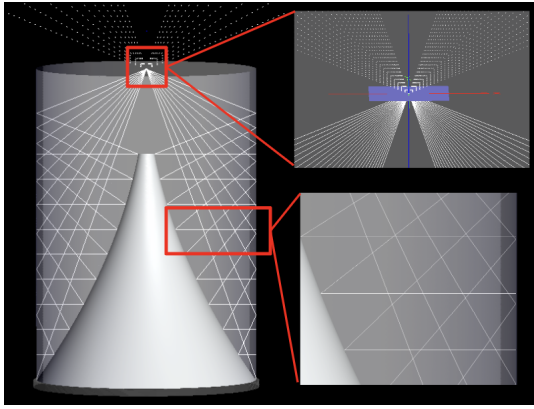
In this high-mass limit of InfraBREAD, two problems arise in the experimental setup. First, whereas the electromagnetic waves in GigaBREAD are emitted coherently with constant phase shift and defined diffraction effects, the electromagnetic waves in InfraBREAD are emitted incoherently with varying phase differences resulting in an imprecise focal spot. This effect has been observed in other reflector experiments using dish antennas [8]. The incoherence arises from the nonzero velocity of incoming axions, which causes the photon emission angle to deviate from the orthogonal direction of the cylinder, resulting in a smeared focal spot larger than 1 mm^2 , the size of the SNSPD (see Figure 3 and details discussed in section II). Second, the alignment precision is sensitive to changes on the millimeter scale in this frequency range. Accordingly, slight shifts in the positioning of the reflector and SNSPD during the experimental cool-down could drastically affect the signal sensitivity.

As such, there is a need to investigate an additional optical configuration around the detector in InfraBREAD that accomplishes the following two goals:

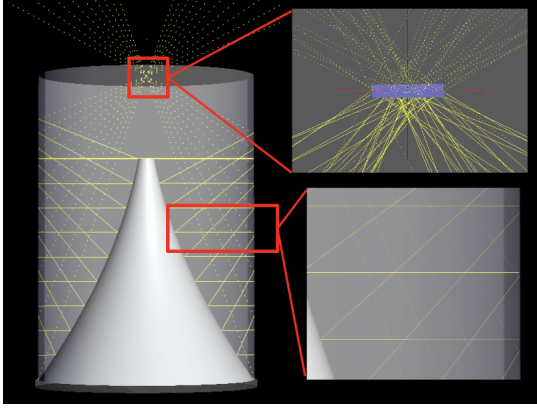
1. captures more rays at this smeared focal spot to increase the signal efficiency of the SNSPD
2. maintains high efficiency of the SNSPD through millimeter-shifts away from the focal spot due to experimental misalignment effects

B. Methodology

Simulations to test the performance of this additional optical configuration were conducted using the ray-tracing software “FRED” by Photon Engineering. In this study, the coordinate system used positions the origin at the location of the precise focal spot at the top of BREAD, with the x-y plane as the focal plane and the z-axis as the optical axis. The simulations considered both the ideal scenario in which photons are emitted perpendicularly from the surface of the cylinder producing a precise focal spot, and the actual scenario in which photons are emitted at within an angular range of 1 mrad from the normal producing an imprecise and smeared focal spot. Given the perpetual motion of the earth around the sun and through the dark matter halo, InfraBREAD



(a) Ideal Scenario



(b) Actual Scenario

Figure 3: Ideal scenario rays are emitted at 0rad from the surface normal making a precise focal spot. Actual scenario rays are emitted at 0.001rad making a smeared focal spot.

will always suffer from finite smearing of the focal spot but the ideal scenario serves as a good baseline approximation. These simulations measured irradiance (power per unit area) absorbed on the detector through various optical configurations and velocity distributions, with thousands of rays randomly distributed along the cylinder and traced. While far fewer electromagnetic waves would be emitted at a time in the real experiment in either the actual or ideal focal spot scenarios, integrating the power over all possible ray source positions in a single run provides an estimate of maximal signal efficiency and the input for an accurate calculation of S/N.

The methodology used to identify an optical configuration and test its performance consisted of two phases:

1. *Phase 1*: in the ideal (precise focal spot) scenario, identify an optical configuration which captures 100% of emitted photon rays at the origin
2. *Phase 2*: in the actual (imprecise focal spot) scenario, analyze the irradiance efficiency, incident ray count, and incident angular statistics on the

SNSPD within this optical configuration versus the SNSPD alone through misalignment shifts.

In the rest of this paper, I will first provide deeper analysis of the halo velocity distribution inside BREAD for the ideal and actual scenarios. Then I will discuss optical configurations tested in phase 1 of the methodology. Lastly, I will discuss the Phase 2 results of one novel optical design called the “Paraboloid to Winston Cone (PTW) Configuration” that has promising potential to fix these problems and its implications.

II. VELOCITY DISTRIBUTIONS IN IDEAL AND ACTUAL SCENARIOS

A. Ideal Scenario

In the ideal scenario where photons are emitted perpendicularly to the outer cylinder, the photon trajectories form a precise focal spot at the origin and then exit the BREAD configuration within a uniform, azimuthally symmetric angular section bounded by ϕ_{min} and ϕ_{max} (Fig 4). Any photon emitted from the base of the outer cylinder will trace the path at ϕ_{min} and any ray that is emitted from the cutoff height of the parabolic reflector will trace the path at ϕ_{max} . A directional analysis entity in FRED computed $\phi_{min} = 19.35^\circ = 0.338\text{rad}$ and $\phi_{max} = 67.95^\circ = 1.186\text{rad}$. Thus, any optical configuration can only pass Phase 1 if it captures all of the rays between ϕ_{min} and ϕ_{max} in the ideal scenario.

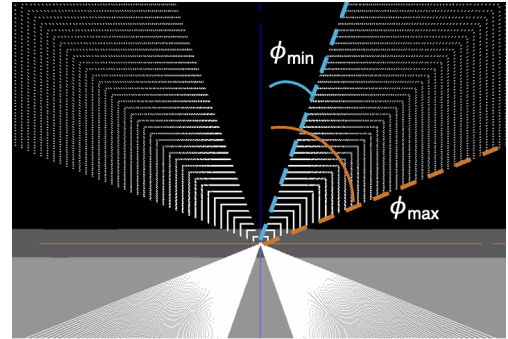


Figure 4: ϕ_{min} and ϕ_{max}

B. Actual Scenario

Currently, the Standard Halo Model (SHM) of dark matter posits that dark matter permeates throughout space and often encompass galaxies such as our own in a gas-like “halo” structure [9]. Variations of the SHM predict different dark matter densities, spatial distributions, and velocities [10]. In these simulations we assume the halo is a spherically symmetric, isothermal gas-like structure with axions moving in different directions with

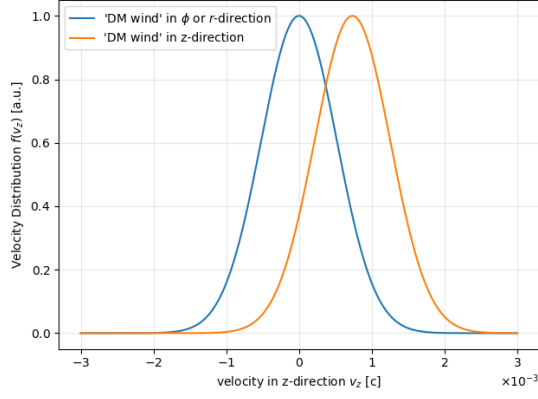


Figure 5: Gaussian halo velocity distribution. Example shown for wind in the z-direction.

nonzero velocity as the earth passes through it. In order to satisfy the $E_{||} = 0$ boundary condition on the metal cylinder and momentum and energy conservation, incoming non-relativistic axions and dark photons will have an outgoing angle α from the surface equivalent to the magnitude of their parallel velocity component, $\alpha \approx |\frac{\mathbf{p}_{||}}{m_{DM}}| \approx 10^{-3}\text{rad}$ (see [11] for a derivation). Depending on the rotation of the earth and its position around the sun, the direction of the dark matter “wind” could change throughout a day or year. These wind directions can be categorized into three cases for BREAD: (a) wind in the z direction, (b) wind in the radial direction, or (c) a “complete” superposition of the two. For simplicity, we assigned any radial wind to a wind in the x-direction.

These conditions of the SHM yield a Gaussian velocity distribution

$$f_{SHM}(\mathbf{v}) = \frac{1}{(2\pi\sigma_{SHM})^{3/2}} \exp\left(-\frac{(\mathbf{v} - \mathbf{v}_{lab})^2}{2\sigma_{SHM}^2}\right) \quad (2)$$

with the standard deviation $\sigma_{SHM} = 156 \text{ km s}^{-1}$ and expectation value $\mathbf{v}_{lab}$ depending on the wind direction and severity [10]. In the case of no wind, $\mathbf{v}_{lab} = 0$ so that the expected parallel velocity component (and outgoing emission angle) is 0. In the case of wind in any direction, $\mathbf{v}_{lab} = 220 \text{ km s}^{-1}$ as set by the SHM. The Gaussian distribution for a “complete” superposition of winds is then determined by multiplying together velocity distributions of the multiple wind directions. These Gaussian distributions can also be used as probability density functions to assign weights to traced rays in FRED. For example, in the case of z-wind, a photon emitted at an angle of 1 mrad in the z-direction will be assigned a weighted power of 1 whereas a photon emitted at any other angle will have some power less than 1 (see Figure 5).

The three cases of stream direction— v_z , v_x , complete v —will have different effects and severity on the smearing of the focal spot. A Python script was made for each scenario and randomly generated 10^6 normalized rays on the inner surface of the cylinder initially all pointing in

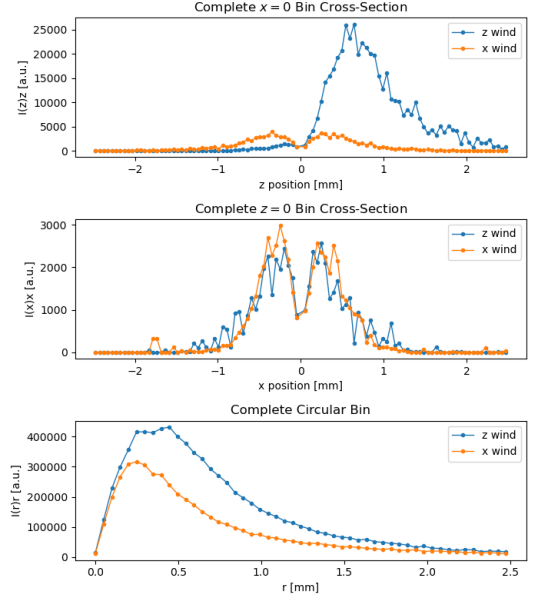


Figure 6: Smeared focal spot for complete v in vertical x-z plane. See Figure 7 (c) for the same irradiance distribution viewed in the horizontal plane.

the direction of the surface normal vector. The rays were then adjusted to the corresponding power weights and rotated according to the direction of the given wind case. In the v_z only case, each ray was randomly rotated to an angle between $[-0.003 \text{ rad}, 0.003 \text{ rad}]$ only in the z-direction. Then to simulate the effects of z-wind, the peak of the Gaussian velocity distribution was shifted to $0.001c$ and the rays emitted at 0.001 rad from the surface normal received the highest power; to simulate no wind, the Gaussian distribution was not shifted and rays that were emitted at 0 rad perfectly perpendicular to the surface received the highest power. In the v_x only case, the same procedure was used except the rays were only rotated in the positive x-direction in the case of x-wind. The “complete” case incorporated a superposition of both x and z rotations and a multiplication of weights, either with dominant z-wind or dominant x-wind; with dominant z-wind, the v_z Gaussian velocity distribution was centered at $0.001c$ and the v_x distribution at 0, and vice-versa for dominant x-wind. A $2.5 \text{ mm} \times 2.5 \text{ mm}$ horizontal analysis surface in FRED centered at the origin in the x-y plane measured the total irradiance across a grid of small square bins of area 0.0016 mm^2 . Figure 6 shows the vertical smearing of the focal spot specifically for the complete v case, and Figure 7 shows the horizontal smeared focal spots for all of the wind cases.

It is evident that the worst case scenario is the complete distribution with dominant wind in the z-direction. In this scenario, the focal spot is smeared across 3mm in the horizontal plane and 2mm in the vertical plane. The maximal irradiance is also approximately 50% reduced from the total possible irradiance as a result of the su-

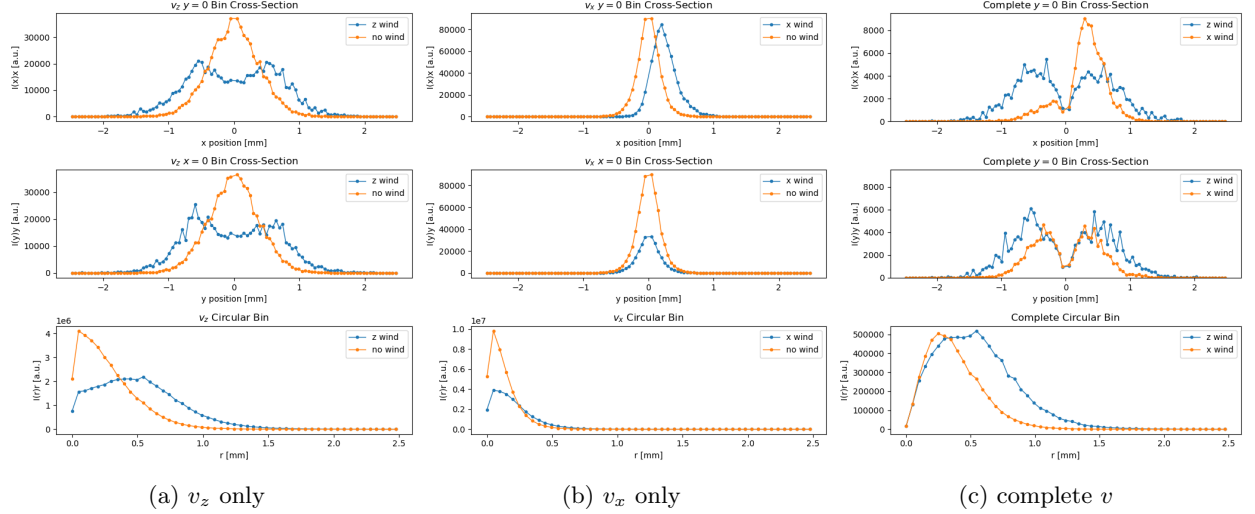


Figure 7: For each wind scenario, the resulting smeared focal spot irradiance distributions in the horizontal x-y plane are shown. Three different binning methods were used: a cross-section along the x-direction of a single row of grid bins, a cross-section along the y-direction also with grid bins, and a re-parameterization of bins in terms of concentric circles emanating out from the center of the analysis surface

perposition of wind directions, which is consistent with the predictions in Figure 6 of [6]. When comparing Figure 6 of this paper and Figure 7 (c), notice also that the complete z-wind case has azimuthal asymmetry around the z-axis but is asymmetric across the x-y plane. To prepare for the worst case scenario, the complete z-wind velocity distribution was used to trace the rays in FRED for the actual focusing scenario.

III. PHASE 1: TESTING OPTICAL CONFIGURATIONS

While multiple optical configurations were tested, two particular configurations were analyzed most closely. These configurations used a combination of reflector and lensing optics in attempt to focus outgoing rays from the smeared focal spot onto an area of 1 mm^2 . This analysis utilized three different types of optical component: compound parabolic concentrators (also known as Winston Cones), convex lenses, parabolic reflectors.

A. Lens + Winston Cone Configuration

The first idea was to use a convex lens that refracts outgoing rays vertically parallel to the z-axis into a Winston cone.

1. Lens and Winston Cone Optics

Developed by physicist Roland Winston, a Winston cone is an off-axis parabola of revolution which focuses electromagnetic waves from an entrance opening into a

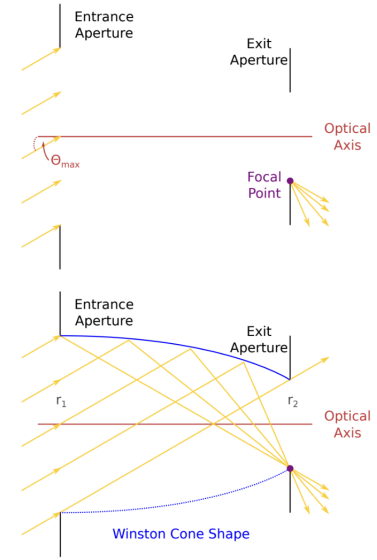


Figure 8: Construction of a Winston Cone. Figure obtained from [14].

smaller exit opening [12]. They are exploited in many non-imaging optics applications such as in photovoltaics [13]. The construction of a Winston cone is parameterized by r_1 , the radius of the entrance hole, r_2 the radius of the exit hole $r_2 < r_1$, and θ_{max} , the tilt of the off-axis parabola and the maximum photon incident angle of acceptance (Fig. 8) [14].

Any photon that enters the Winston cone outside of $[-\theta_{max}, \theta_{max}]$ will be reflected back out of the entrance hole and not pass through the exit hole. At the threshold angles of $\pm\theta_{max}$, the incoming rays will be re-

flected to the focal spot of the opposite parabola at the edge of the exit hole. This Winston cone threshold angle is given by

$$\sin(\theta_{max}) = \frac{r_2}{r_1} \quad (3)$$

such that as the difference $r_1 - r_2$ increases, $\theta_{max} \rightarrow 0$. When $r_2 \ll r_1$, θ_{max} rapidly decreases, limiting the range for angular acceptance of incoming rays to those that are nearly parallel to the imaging axis (Figure 10 shows how θ_{max} varies for the chosen Winston cone parameters for InfraBREAD). The length is given by

$$l = \frac{r_1 + r_2}{\tan(\theta_{max})} \quad (4)$$

and in this same limit where $r_2 \ll r_1$, $\tan(\theta_{max}) \approx \theta \ll 1$ so the Winston cone becomes very long. The idea is to use a Winston cone aligned with the z-axis and $r_2 = 0.5\text{mm}$ so that the signal focus onto the 1mm^2 area of the SNSPD. Now an additional optical component is needed below the Winston cone to send outgoing rays into its entrance aperture.

The first thought was to use a convex lens, but the effects of spherical aberration must also be considered given the sensitivity of the Winston cone threshold angle. Figure 11 shows the geometric parameters of a spherical convex lens. Normally, the relationship between focal length f and lens radius R is given by the lensmaker equation

$$\frac{1}{f} = (n - 1) \left[\frac{2}{R} + \frac{(n - 1)d}{nR^2} \right] \quad (5)$$

assuming that the radii of the front and back lenses are the same. Spherical aberration perturbs this focal length.

One approach to mitigating the spherical aberration effects is to use a parabolic lens. When compared to spherical lenses in FRED, parabolic lenses resulted in greater parallelity between refracted rays, and a nonzero width d improved this effect even more. In addition, lenses with indices of refraction higher than glass, such as silicon ($n = 3.935$) or gallium arsenide ($n = 4.24$), can reduce spherical aberration since the greater intensity of refraction allows for the lens to be thinner (see Figure 9).

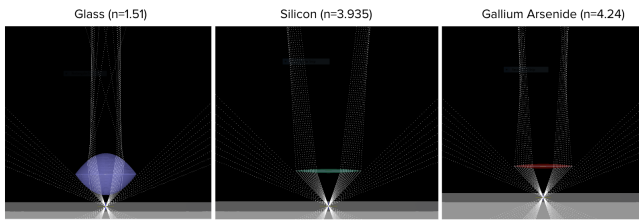


Figure 9: Photon refraction from lenses with different n

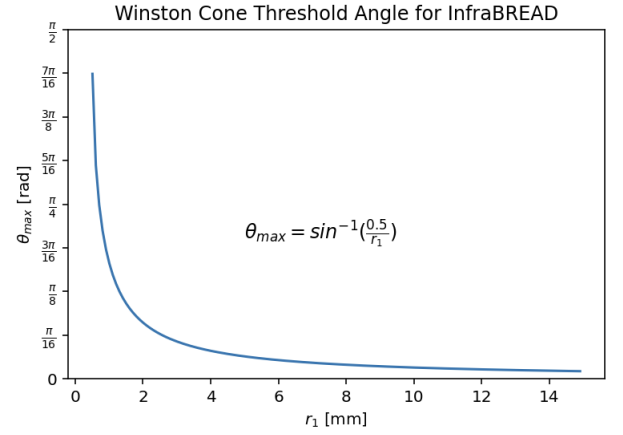


Figure 10: Winston cone threshold angle θ_{max} as a function of r_1 for $r_2 = 0.5$.

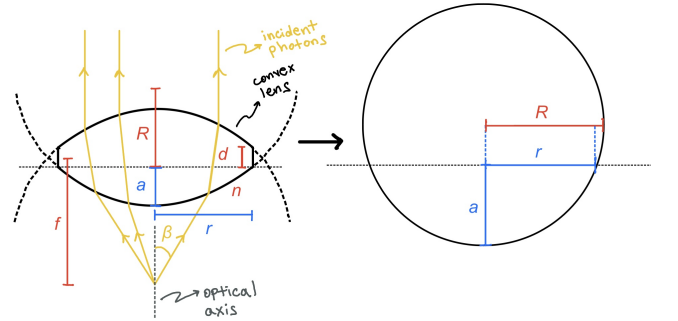


Figure 11: Spherical convex lens schematic. Red variables appear in lensmaker equation. In blue, r is the radius of the lens cut from a sphere of radius R , and a is the width of the lens. The right diagram shows the large a when $r \approx R$ for the fast lensing in BREAD, leading to spherical aberration.

2. Analysis of Lens + Winston Cone Configuration

By placing the lens at the height of its focal length, the desired outcome of this optical configuration would be to refract all of the outgoing rays vertically within the Winston cone threshold angle so that they are all focused within a 1mm^2 area. However, even with a parabolic lens of higher index of refraction than glass, a closer mathematical analysis and FRED simulations reveal that this configuration cannot physically achieve this desired effect due to the fast-imaging nature of BREAD and the corresponding small angular acceptance of the Winston cone. Three quantitative arguments demonstrate that this setup will always suffer from spherical and chromatic aberrations:

Firstly, the low path traced by the photon at ϕ_{max} in the ideal scenario sets a stringent geometric constraint if this optical configuration is to capture 100% of the outgoing rays. The constraint enforces that, at a minimum,

the relationship between r and f must satisfy

$$f = r \tan\left(\frac{\pi}{2} - \phi_{max}\right) \quad (6)$$

For any size of lens, this constraint enforces that $f < r$. Therefore, this setup will have a very low “f-number” N , which is the ratio between the focal length and the diameter of the lens,

$$N = \frac{f}{D} = \frac{f}{2r} = \frac{1}{2} \tan\left(\frac{\pi}{2} - \phi_{max}\right) = 0.202 < 1 \quad (7)$$

That $N < 1$ means the focusing requires extremely fast lensing optics to refract the rays to be parallel with the z -axis. Since $R \propto f$ and since $f < r$, the longitudinal radius r of the lens will be comparable to the full radius R of the proxy sphere or parabola from which the lens is cut, and therefore each lens in this setup will have a large cross-section distance a of this proxy sphere or parabola (see Figure 11). The combination of this large a with $f < r$ results in a significant difference in curvature, or second order approximation, of the surface at the center of the lens at its edges, meaning that incoming rays will intersect with the center of the lens at significantly different angles from those on the lens edges.

Second, traditional optics equations like the lensmaker equation use the paraxial approximation, which assumes that the angles of refraction from the lens are close to the optical axis so $\beta \approx 0$ and therefore $\beta \approx \sin(\beta)$. This approximation yields the relation $\frac{1}{s_o} + \frac{n}{s_i} = \frac{n-1}{R}$ from which one can derive the lensmaker equation, where s_o is the object distance, s_i is the image distance, and n is the index of refraction of the lens. However, the paraxial approximation does not apply to the fast lensing of BREAD where $\phi_{max} \gg 0$. Therefore, the next to leading order term in the expansion $\sin(\theta) = \theta - \frac{\theta^3}{3!}$ becomes dominant so the approximation is no longer linear and this adds a nonlinear term to the above relation between s_o, s_i, n and R which depends on r ,

$$\frac{1}{s_o} + \frac{n}{s_i} = \frac{n-1}{R} + r^2 \left[\frac{1}{2s_o} \left(\frac{1}{s_o} + \frac{1}{R} \right)^2 + \frac{n}{2s_i} \left(\frac{1}{R} - \frac{1}{s_i} \right)^2 \right] \quad (8)$$

As a result of this nonlinearity, effects of spherical aberration will be dominant and the assumption that a lens placed at the height of its focal spot will send rays parallel to the optical axis no longer holds.

Third, the exit aperture of the Winston cone fixed at $r_2 = 0.5\text{mm}$ in combination with the spherical aberration effects from large r of the lens makes the desired irradiance capture not possible. In order to achieve complete ray capture, it must be that $r_1 \geq r$ and r_1 should be minimized whenever possible to maximize θ_{max} . However, given the optical geometric constraints of InfraBREAD discussed above, the Winston cone will be in the limit where $r_1 \gg r_2 = 0.5\text{mm}$, which forces θ_{max} to be very small as shown in Fig. 10. One could instead use a lens with larger focal length and therefore smaller a , but this

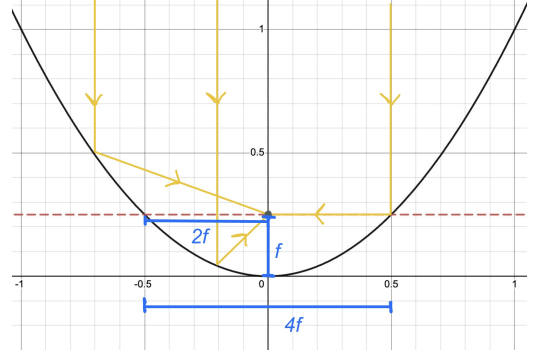


Figure 12: Parabola optics for a given focal length f

would reduce the degree to which the rays are refracted and require a larger Winston cone entrance radius r_1 , which then decreases the threshold angle even further.

Beyond spherical aberration, this setup will suffer from chromatic aberrations. Different wavelengths of photons will refract at different angles through a lens material with a fixed index of refraction and therefore the SNSPD will not have a constant efficiency across the spectrum of wavelengths. Hence, lensing is not an ideal tool for the broadband nature of BREAD and instead, an optical configuration relying almost entirely on reflector optics would be more ideal. Other configurations and modifications of this setup were tested and are highlighted in Appendix A.

B. Paraboloid to Winston Cone (PTW) Configuration

A revised optical configuration was identified using only reflector optics. Instead of a convex lens, it uses a parabolic reflector to send rays perfectly parallel to the optical axis and into the Winston cone.

1. Parabola Optics

Every parabolic reflector has an associated focal point, such that any incoming wave parallel to the optical axis will be reflected to that focal point. For an arbitrary parabola of the form $y = ax^2$, the focal length is given by $1/4a$. The cross section of the parabola at the focal length height has a width of $4f$ (Fig. 12).

2. Analysis of PTW Configuration

We can take advantage of the optics of a parabola by situating its focal spot with the focal spot from BREAD at the origin. Almost acting like a reverse Winston cone, one can cut an entrance hole into a paraboloid at the height of its focal length and position the paraboloid on the focal plane at $z = 0$. Any ray going through this precise focal spot in the ideal scenario that comes in contact

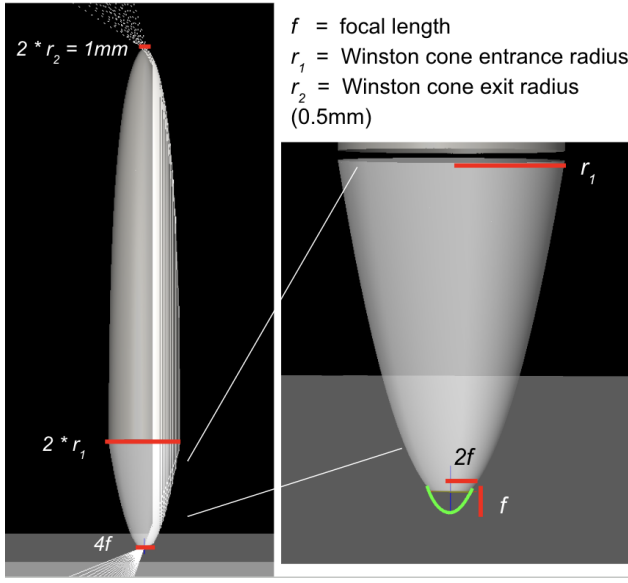


Figure 13: PTW Configuration

with the paraboloid will then be reflected perfectly parallel to the z-axis and thus enter the Winston cone at an incident angle of 0° , so that every ray will pass through the exit aperture (see Figure 13). The entrance radius r_1 of the Winston cone can be made equal to the radius of the exit hole of the parabola for optimal alignment.

Another benefit of this configuration is that the entire setup can be parameterized by a single variable, the focal length of the parabola f . Instead of defining the paraboloid in terms of some arbitrary coefficient on the quadratic coefficient a , the equation in the BREAD global coordinate system can be re-written as $z = 1/(4f)x^2 - f$. In this framework, the entrance radius of the parabola is $r_0 = 2f$, and diameter $D = 4f$, and the exit radius of the paraboloid (equivalently the entrance radius of the Winston cone) r_1 can be determined from focal length depending on different cutting models (two are discussed in section IV and Appendix B). IR Labs can manufacture both the Winston cone and custom paraboloid.

IV. PHASE 2: PTW CONFIGURATION

A. PTW Models

Now that a functioning optical setup—the PTW configuration—was identified and captured 100% of outgoing rays at the origin in the ideal focal spot scenario, it was analyzed in the actual scenario to determine how well it maximizes irradiance at the origin and through misalignment shifts relative to the SNSPD alone without any extra optical components. Two models with different methods of setting the exit hole size of the paraboloid were simulated, one optimizing the Winston cone thresh-

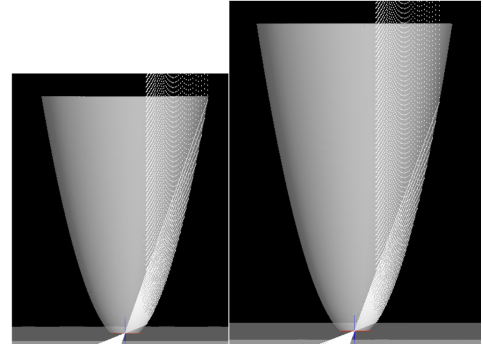


Figure 14: Low cut (left) and High cut (right) PTW Models

old angle at the origin, and the other optimizing the irradiance longevity from misalignment shifts. They are referred to as the “low-cut” and “high-cut” models, respectively.

The construction of these two models depends on where the highest ray exiting BREAD intersects with the paraboloid; our goal is to capture all of the outgoing rays and if we capture this highest one, then we capture all of the ones below it. Given that we cannot define an exact position and angle of this highest ray in the actual incoherent focal spot scenario, we can use the ideal scenario as an approximation, with the highest ray corresponding to the one tracing out ϕ_{min} (see Figure 4). The low-cut model cuts exit aperture of the paraboloid at the height where it intersects with the line traced out by ϕ_{min} when centered at the origin, minimizing the entrance aperture of the Winston cone and therefore maximizing the threshold angle given the constraints of BREAD. Alternatively, the high-cut model cuts the height of the paraboloid where the line given by ϕ_{min} intersects with the paraboloid when it is maximally shifted away from the origin at a horizontal offset of $x_{max} = r_0 = 2f$ (after this distance it completely misses the focal spot in the ideal scenario). See Appendix B for the mathematical derivation of these two models.

B. Simulation Setup

Simulations of the performance of the PTW configuration in FRED traced rays from the worst case velocity distribution inside of BREAD, which was the complete distribution with wind predominantly in the z-direction. In each simulation, 10^5 rays were traced with a wavelength of $1.013 \mu\text{m}$. Rays were randomly distributed on the inner wall of the cylinder between the base of BREAD and the cutoff height of the parabolic reflector, such that all the rays emitted rays should exit the BREAD setup. Scripts were written in FRED in the WinWrap language to automate and record data from shifts in the setup. At each given shift value, the measured observables of the simulations on the SNSPD in both the PTW setup

and the SNSPD alone were irradiance efficiency I , incident ray count C , and average, minimum, and maximum incident angle β .

The irradiance efficiency I was calculated by creating an “irradiance spread function” in a FRED Analysis Results Node (ARN) through a full raytrace. The calculation was given by,

$$I = \frac{\iint_{surf} P_{abs} dA}{P_{tot}} \quad (9)$$

where the absorbed power P_{abs} was integrated over the grid bin size of $dA = 0.0001 \text{ mm}^2$ on the 1 mm^2 SNSPD surface and divided by the measured total power P_{tot} of rays exiting the BREAD cylinder. Due to small errors in coarse grain resolution from the surfaces in FRED and a fraction of emitted rays either getting trapped between walls or underneath the central parabolic reflector, P_{tot} turned out to be less than the total weighted sum of all of the simulated rays. Placing a circular analysis plane of radius 201mm that completely covered the top of the cylinder, it was determined that $P_{tot} = 3495.126$ [a.u.] which is approximately 75.16% of the total sum of weighted power.

The ray count C was determined by a function in FRED that counts number of incident rays on a surface.

The incident angle statistics were calculated in the following way: the x, y, and z components of each incident ray (r_x, r_y, r_z) were transformed into the coordinate system of the detector, with an angle from the polar z-axis of

$$\beta = \pi/2 - \arctan \left(\frac{r_z}{\sqrt{r_x^2 + r_y^2}} \right) \quad (10)$$

The velocity distributions inside BREAD show that the maximum spread of the focal spot occurs roughly within the horizontal range $x \in [-1.5, 1.5] \text{ mm}$ with the tails falling off to zero around a maximum of $\pm 2 \text{ mm}$, and within the vertical range $z \in [0, 2] \text{ mm}$. As such, the misalignment simulations considered shifts in this full range, with a step size of 0.1mm. In addition, PTW configurations with different focal lengths were tested from a focal length of 0.25mm to 1mm with a step size of 0.125mm. The focal length 0.25mm was chosen as the lowest limit because it makes the paraboloid entrance hole diameter $D = 1 \text{ mm}$ (the same as the SNSPD), so it serves as a control against the SNSPD by itself. Moreover, since any wind in the z-direction will be azimuthally symmetric for a given horizontal cross section of the focal spot as shown in Figure 7 (c), the simulations of shifts in x only considered shifts in the positive x-direction from 0mm to 2mm to save computation time. The z-wind does not have polar symmetry, however, so simulations for shifts in z considered a full sweep from -2mm to 2mm.

C. PTW High Cut vs. Low Cut Results

The results in Figure 15 show a full sweep through misalignment shifts in x and z and various focal lengths for both the high cut and low cut PTW models. Both models show nearly identical trends through misalignment shifts, meaning that they exhibit relatively the same degree of longevity in irradiance capture. However, for all focal lengths, each one of the PTW low cut sizes exhibit a higher irradiance efficiency both at the origin and through shifts than each of the corresponding PTW high cut sizes. Figure 16 shows the maximum efficiencies attained by each focal length for both the high and low cut models through the entire sweep of misalignment shifts, and on average the low cut model achieves around 5% greater efficiency than the high cut model. Note that while this trend in maximum efficiencies shows the start of a slight sinusoidal behavior for larger focal lengths, any PTW configuration with $f > 1 \text{ mm}$ is infeasible to manufacture given that the Winston cone length is already $l = 283.40 \text{ mm}$ at $f = 1 \text{ mm}$.

The highest achieved efficiency out of any of these simulated models is the PTW low cut model at $f = 0.5 \text{ mm}$ (bright green), which sets the corresponding Winston cone entrance aperture $r_1 = 5.83 \text{ mm}$ and yields a Winston cone threshold angle $\theta_{max} = 0.086 \text{ rad} = 4.92^\circ$. This particular configuration has a paraboloid entrance diameter $D = 2 \text{ mm}$, which is twice the size of the SNSPD. It is at this size that the inverse relationship between the incident ray capture area and Winston Cone threshold angle is optimized. The PTW low cut $f = 0.5 \text{ mm}$ achieves a maximal efficiency of 60% for shifts in x and 74% for shifts in z. We can now compare this optimal PTS low cut $f = 0.5 \text{ mm}$ model to the SNSPD alone, and the simulations hereafter only trace through the focal lengths up to 0.5mm.

D. Comparison to SNSPD Alone

Simulation results comparing PTW low cut models of focal lengths 0.25mm, 0.375mm, and 0.5mm with the SNSPD alone are shown in Figure 17.

The PTW configurations of $f = 0.375 \text{ mm}$ and $f = 0.5 \text{ mm}$ maintain a higher irradiance efficiency through all misalignment shifts in x. Through misalignment shifts in the negative z-direction, the PTW configurations increase the irradiance efficiency from the SNSPD alone significantly, but in positive z-shifts the SNSPD alone achieves a slightly higher efficiency. The shift locations at which the maximum efficiency values occur for each plotted curve aligns with the velocity distribution for complete z wind. In the horizontal plane, the distribution is smeared evenly on both sides of the z-axis so a detector of semi-aperture 0.5mm would only achieve a maximum of roughly 50% efficiency at the origin (see again Figure 6 of [6]); in the vertical plane, the velocity distribution is shifted towards the positive z-direction with the peak

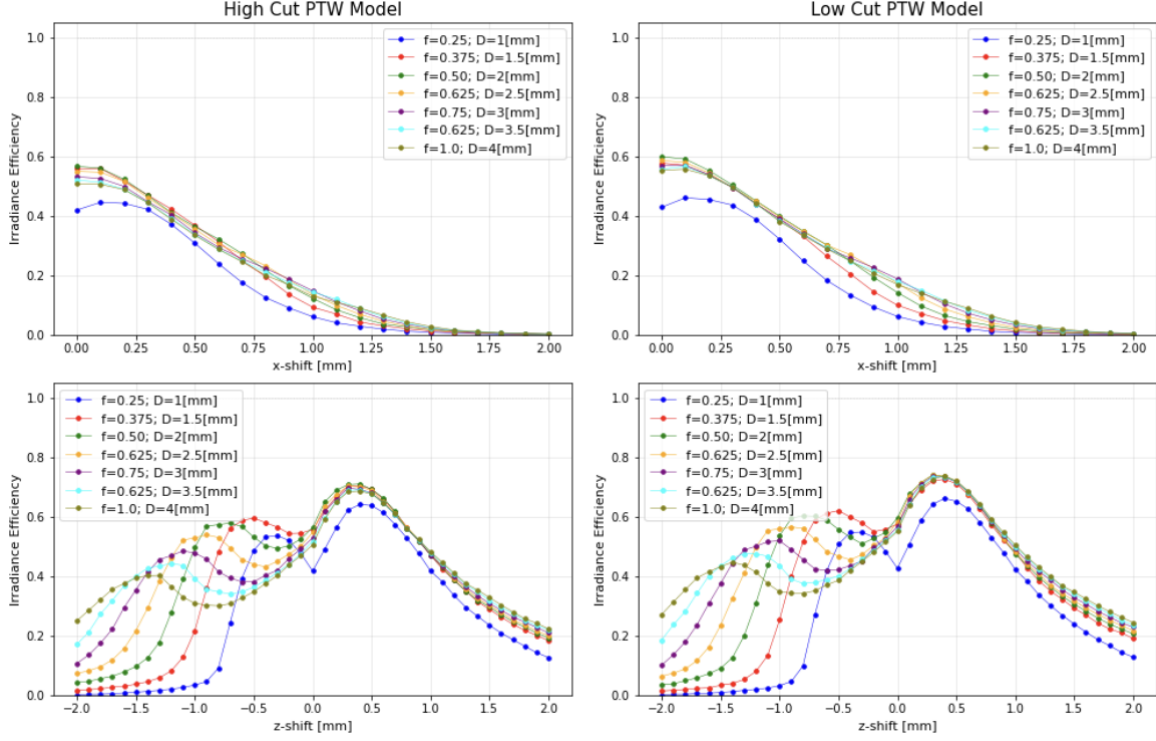


Figure 15: Results for PTW High Cut vs. Low Cut through x and z-shifts

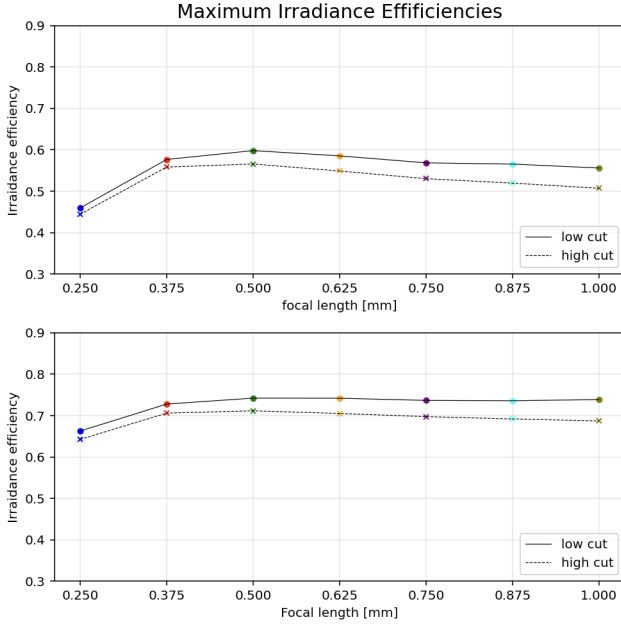


Figure 16: Maximum irradiance efficiencies for PTW Low vs. High cut. The top plot shows these values through x-shifts, and the bottom through z-shifts.

near $z=0.5\text{mm}$, which is also the general region containing the locations of the max efficiency for both the PTW configurations and the SNSPD alone.

The optimal PTW low cut $f = 0.5\text{mm}$ configuration increases the efficiency of the SNSPD alone by 8% at the origin and maintains this relatively constant efficiency increase through all x-shifts. In positive z-shifts, the PTW low cut $f = 0.5\text{mm}$ demonstrates a lower efficiency than the SNSPD alone but only at a maximal deficit of at 7% at $z = 0.7\text{mm}$, whereas in negative z-shifts it produces a maximal gain in efficiency of 55% at $z = -0.8\text{mm}$.

One advantage that the SNSPD alone has over the PTW configuration is in its incident angular range. The detector in the PTW configuration consistently demonstrates an angular range (the difference between maximum and minimum incident angle) of approximately $\pi/2$ at all locations, meaning that the incident angles hit the SNSPD at all possible angles. This results from the conservation of area in phase space, since any decrease in the area of focal spot smearing will result in an increase in angular range. The SNSPD alone does not apply any optical transformations of phase space so it experiences the normal angular range of incident angles from the smeared focal spot of approximately 0.85 rad, and this value even decreases on the fringes of z-shifts.

E. Further simulations and comparisons

a. Double-Bump Feature of PTW. The peculiar double bump feature of the PTW irradiance efficiency distribution through z-shifts prompts deeper investiga-

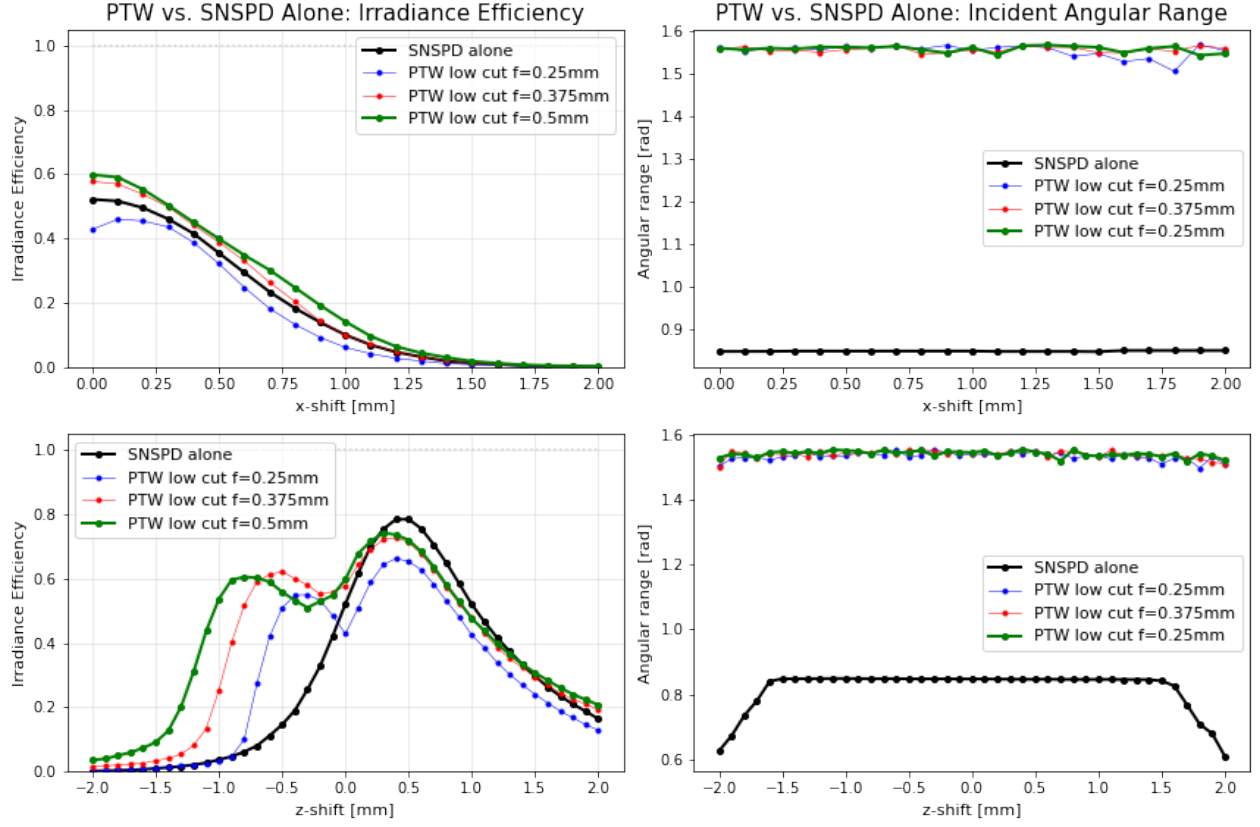
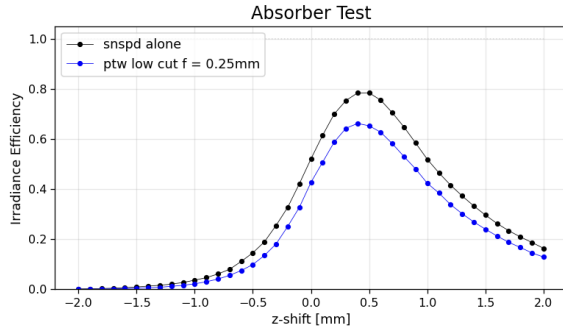


Figure 17: Results for PTW Low Cut vs. SNSPD Alone

Figure 18: PTW low cut $f = 0.25\text{mm}$ vs. SNSPD alone with absorber. Results show blue curve uniformly bounded by black curve.

tion. In the region between $[-1, 0]$ in z -shifts in Fig. 17, the PTW low cut $f = 0.25\text{mm}$ configuration exceeds the SNSPD alone in both irradiance efficiency and incident ray count. From first glance, this would appear physically impossible given that the $f = 0.25\text{mm}$ model (used as a control) has a paraboloid entrance has an area $\pi 0.5^2 < 1\text{mm}^2$, so we would expect the blue curve to be uniformly bounded by the black curve. A follow-up simulation placed a horizontal absorber at the height of the bottom of the paraboloid and covered everything ex-

cept the entrance hole of the paraboloid. The absorber eliminated any extra reflections off the external surface of the PTW configuration from rays outside of the 1mm^2 . This results of this test shown in Figure 18 produce the expected distributions.

Therefore, we can deduce that in the z -range of $[-1, 0]$, the smeared rays outside of the entrance hole of the parabola reflect off the outside of the paraboloid back down into BREAD and make their way into the PTW configuration. This is not a flaw in simulation but rather a unexpected benefit of the PTW setup. Indeed, notice in Figure 17 that each of the focal length parametrizations have this double bump feature with the peak of the bumping shifting to the negative z -direction as the focal length and corresponding angle of the tangent plane of the parabola increases. This suggests that the strong increase in efficiency of the PTW configuration for low z -shift values is largely due to this secondary external reflection. This external reflection will be explored in further simulations.

b. Off-axis Rotations. These simulations of misalignment shifts all assumed that the PTW configuration and SNSPD alone were always aligned with the optical z -axis. It is also worth considering rotational misalignments in the optical axis of system through these shifts in position. The effects of off-axis rotations of 5° , 10° , and 15° were simulated through shifts in x and z . To reduce

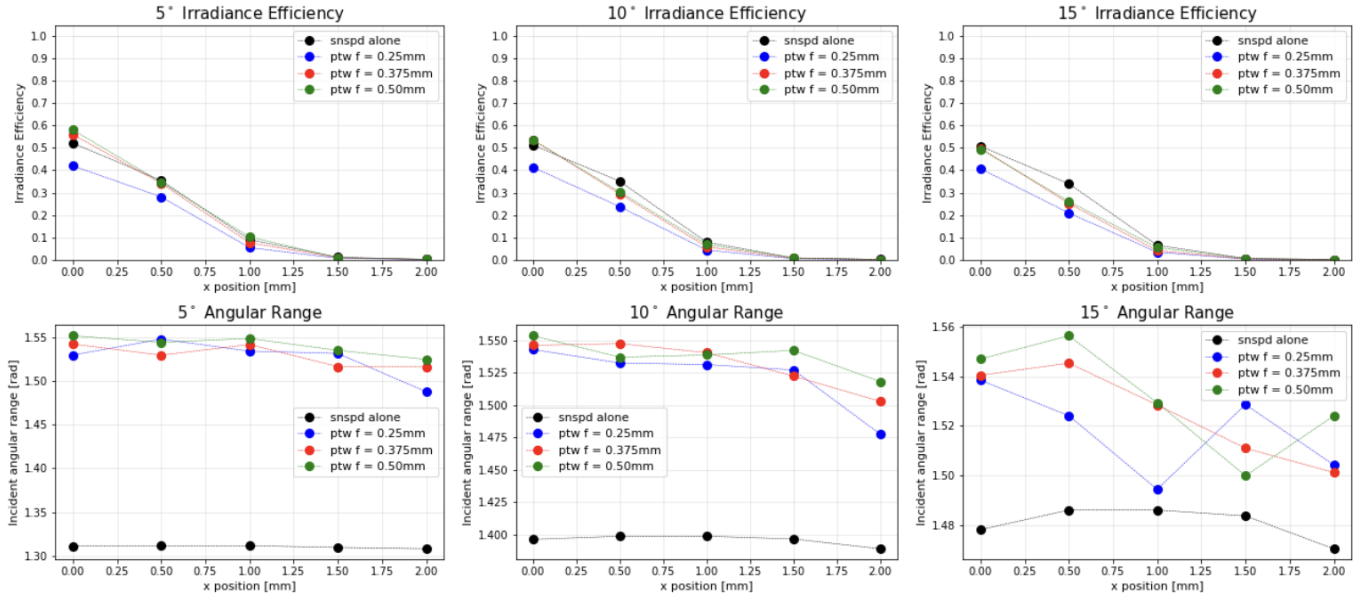


Figure 19: Irradiance efficiency and incident angular for off-axis rotations of 5° , 10° , 15° . Only x-shifts are shown.

computational run-time, the shift step size was increased to 0.5mm for the x-shifts and 1mm for the z-shifts. The trend in irradiance efficiency for each rotation in both x and z shifts follow the same distribution shape as in Figure 17 with 0° rotation, which means that the system is largely insensitive to rotations and more sensitive to shifts in position. Because of this result, only the x-shift plots for irradiance efficiency and incident angular range are shown in Figure 19. Notice that at small rotations, the PTW configurations outperform the SNSPD, however with larger rotations the SNSPD attains a higher efficiency. This makes sense in the context of a simple flux argument where a rotated square with incident irradiance $\int P_{abs} \cos(\beta) dA$ would yield a greater efficiency than a ray traveling through a rotated parabola and a Winston cone.

One noteworthy feature of these off-axis rotations through shifts in x in particular is their effect on the incident angular range. As the degree of the off-axis rotations increases, the angular range values across the PTW configurations and the SNSPD alone begin to wobble. At 15° of off-axis rotation, the PTW angular ranges resemble sinusoidal behavior through shifts in x, with the frequency of oscillation decreasing as focal length increases, with the SNSPD showing the lowest frequency. The phase space explanation for this behavior is still unknown and will be explored in future simulations.

c. SNSPDs of Larger Size. Single-photon counting detectors are an active and rapidly developing field of physics research. Hence, it is also important to consider the future potential for an SNSPD of larger surface area, or alternatively an array of multiple 1 mm^2 SNSPDs. Another round of simulations analyzed the performance of SNSPDs of linearly increasing surface area from 1 mm^2

Table I: Maximum irradiance efficiencies for SNSPDs of various sizes through misalignment shifts

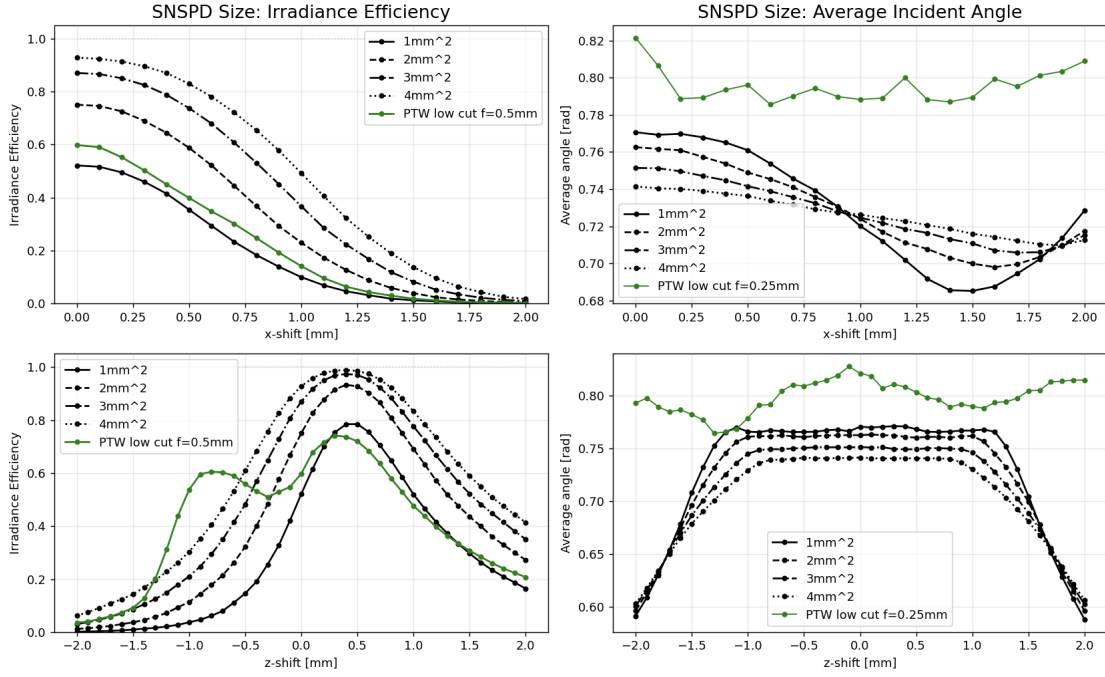
SNSPD size [mm^2]	x-shift max. irradiance eff.	z-shift max. irradiance eff.
1	52%	78%
2	75%	93%
3	87%	97%
4	93%	99%

to 4 mm^2 shown in Figure 20.

Evidently, the efficiency of the SNSPD quickly increases as the surface area increases. Even with just doubling the size of the SNSPD, the peak efficiency increases by 23% for x-shifts and 15% for z-shifts. Increasing the surface area of the SNSPD makes much greater improvements to efficiency than does the best additional optical configuration identified thus far (PTW low cut $f = 0.5 \text{ mm}$), which only improves the efficiency by a maximal amount of 8% in x-shifts. However, in a region of negative z-shifts, the PTW low cut $f = 0.5 \text{ mm}$ attains a higher efficiency than all of these larger SNSPD's because of its double bump structure.

Moreover, increases in SNSPD size greatly improve the incident angular distribution. The same sinusoidal behavior that was observed in angular range from simulations on off-axis rotations is observed for average incident angles on these SNSPD's of varying sizes. The amplitude of oscillation decreases as SNSPD size increases, resulting from the fact that a larger surface area will receive a greater concentration of rays at the average angle value.

However, increasing the surface area of the SNSPD would also increase its background called the dark count

Figure 20: SNSPD of various sizes vs. PTW low cut $f = 0.5\text{mm}$

rate (DCR), which is the signal of other incoming astrophysical particles. Tests by Christina Wang on the SNSPD show that at an optimal bias current of $32\mu\text{A}$ where the difference between the DCR and photon count rate (PCR) is the greatest, a single pixel on the SNSPD has a DCR of approximately 10 counts per second (cps). Assuming that these excess cosmic particles have a uniform density and that the DCR scales linearly with surface area, the entire 1mm^2 SNSPD with 8 pixels then has a DCR= 80 cps, the 2mm^2 has DCR= 160cps, the 3mm^2 has DCR= 240 cps, and the 4mm^2 has DCR= 360 cps.

These tests were performed by shining a bright laser onto the SNSPD directly orthogonal to its surface. The PCR of the actual BREAD experiment will therefore be slightly reduced and not necessarily scale linearly with surface area. Instead, we can get a more accurate estimate of how the PCR scales by using the same Gaussian distribution to compute a weighted sum of incident photons on the SNSPD at a given shift position. At the origin, the SNSPDs then receive a total PCR of 1817.47, 2621.35, 3040.76, 3250.47 for an SNSPD of 1mm^2 , 2mm^2 , 3mm^2 , and 4mm^2 respectively. While one cannot make an accurate S/N comparison between these scaled PCRs and DCRs because the above PCRs are not in units of counts per second, the general trend shows that while DCR scales linearly, the PCR scales logarithmically. As such, there will be a cost to increasing the surface area of the SNSPD because its background grows more rapidly than its signal.

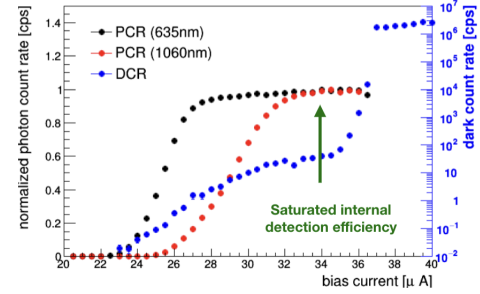


Figure 21: Dark count rate for the SNSPD. Figure obtained from slides of Christina Wang.

V. OUTLOOKS AND FUTURE SIMULATIONS

These simulation results of the optimal PTW configuration with low cut at $f = 0.5\text{mm}$ in comparison to the current 1mm^2 SNSPD alone generally show an improved performance in both maximum efficiency at the origin and longevity. However, as discussed, there is great room for improvement in developing this additional optical configuration. In the future, both modifications to the described PTW setup and potentially investigating entirely different optical configurations should be explored. A combination of any of the below proposals might improve the efficiency even more.

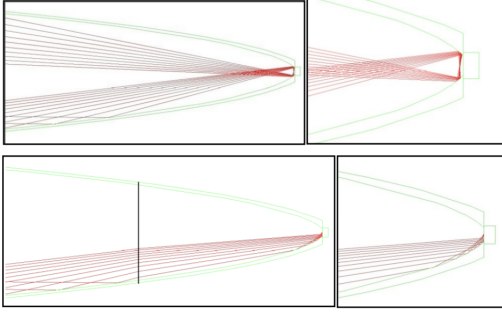


Figure 22: Winston cone with dielectric. Figure obtained from [13]

A. Modifications to PTW Configuration

First and foremost, the most impressive increase in irradiance efficiency of the PTW configuration from the SNSPD alone surprisingly came from the unexpected effect of external surface reflections back into BREAD. Therefore, future simulations should exploit this physical feature and construct an angled disk surrounding the entrance hole of the paraboloid. From existing angular simulation data for a given z -height, one can determine the average incident angle within this range of external reflections and then construct a planar or parabolic reflective surface to send the rays back down into BREAD to be reflected back up more towards the optical axis.

Second, a large factor limiting the irradiance efficiency of the PTW setup is the small Winston cone threshold angle. However, there may be a way to correct for this small angular acceptance inside of the Winston cone cavity itself. Antonini et al. (2013) have introduced the idea of a placing a dielectric material with a high index of refraction inside of the Winston cone before the exit aperture to focus the photon angles towards the optical axis (see Figure 22). While this dielectric would suffer from the same chromatic aberrations discussed in section III, these effects may be negligible given that it is placed at the end of the optical configuration so close to detector.

Third, another way to increase angular acceptance could be to set the Winston cone exit hole radius to be $r_2 > 0.5\text{mm}$. With this increase in θ_{max} and a less constrained phase space, the angular range of rays going through the Winston cone exit hole will decrease. Then, a smaller sized PTW configuration with a final exit radius of 0.5mm could be placed at the top of the larger Winston cone exit hole. That is, this setup would consist of two successive PTW configurations.

Fourth, while the hallmark of the PTW configuration is the construction of the paraboloid entrance hole cut at the focal length height, the velocity distribution in the actual scenario may lend itself to better focusing from cuts above or below the focal length of the paraboloid. If the ray with the highest power from the Gaussian distribution (i.e. the one emitted at 1mrad in the direction

of wind) has a lower intersection angle with the parabola than expected in the ideal scenario, then the entrance hole of the paraboloid should be cut at a lower height. Conversely, if this ray has a higher intersection angle, then the parabola should be cut at a higher height. The latter would be expected for the complete z -wind velocity distribution, where the focal spot is shifted upward in the z -direction. Cutting the paraboloid at a height $h > f$ would also create a larger entrance hole and may therefore increase ray capture.

B. Alternative Optical Configurations

Beyond the PTW setup, another promising reflector optical configuration idea would be to position a parabolic dish reflector facing downward with a 1mm^2 hole cut in its center where the SNSPD would be positioned. Below the parabolic reflector would be a secondary smaller parabolic mirror facing upward that would send reflected rays up into the SNSPD. As shown in Figure 4 of the ideal scenario, any vertical cross section of BREAD will have two angular sectors of electromagnetic waves on opposite sides of the main reflector. This smaller secondary mirror should be placed at a height below the smeared intersection of these two angular sectors in the actual scenario so as to not interfere with their ray paths. Such a configuration would have to be tested again through both Phase 1 and Phase 2 of the simulation methodology.

VI. CONCLUSION

The purpose of this analysis was to identify and simulate an optical configuration with the goal of improving SNSPD performance in two ways:

1. increasing signal power at the origin to re-correct the smeared focal spot
2. increasing signal longevity through misalignment shifts due to experiential cooling or construction errors.

To do so, configurations were first tested in the ideal, precise focal spot scenario (phase 1) and then in the actual, smeared focal spot scenario (phase 2). Phase 1 showed that lens optics suffer from both spherical and chromatic aberrations in BREAD and that reflector optics are better suited. The optical “Paraboloid to Winston Cone” configuration with the low cut model and a focal length of 0.5mm overall accomplished both of the project goals, however there is room for improvement and further simulation analysis to be done.

Through x -shifts in misalignment, the PTW low cut $f = 0.5\text{mm}$ configuration universally achieved greater irradiance efficiency than the SNSPD alone, and increased the efficiency by about 7% through the majority of the

full shift range. Through z-shifts in misalignment, this optimal PTW configuration had a maximal improvement in efficiency by approximately 55% in one region. Although, the large angular range of incident rays on the detector in the PTW configuration (approximately all of $\pi/2$) could suppress this irradiance efficiency [15]. Misalignments from off-axis rotations were also considered and prove to have sub-dominant effects on detector performance in comparison to shifts in position. At small rotations, the PTW configuration does on average outperform the SNSPD alone, whereas at larger rotations the opposite is true. SNSPD's of larger surface area of 1 mm^2 showed the greatest improvement in both irradiance efficiency and incident angular range, however this observed increase in efficiency may be suppressed as the DCR increases with surface area.

Moreover, gaining empirical intuition for the types and severity of misalignments would be important for placing accurate constraints and conclusions from these simulations. If it turns out that experiential shifts from cooling beyond $\pm 0.5 \text{ mm}$ is expected, then it would be worthwhile to further develop and implement an optical configuration like PTW. Otherwise, the simplicity of the SNSPD alone and its potential for increased surface area may be more desirable for InfraBREAD.

Beyond the distinctions between these optical configurations, this study illuminates the complexity of optical focusing required for dish-antenna direct detection experiments searching for high-mass dark matter. Further enhancements and modifications to the performance of single-photon counting detectors, such as increasing their surface area, will be vital and should continue to be explored in future research.

ACKNOWLEDGMENTS

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Appendix A: Appendix A: Other Optical Configurations Tested in Phase 1

Other configurations that were tested in Phase 1 are highlighted below.

a. Stacked Lenses

While a single lens alone has shown to be incapable of achieving the desired focusing effect in InfraBREAD, a series of stacked lenses, all with $f > r$, was analyzed.

Although each of these lenses individually would not refract the incoming photons at a strongly enough to make them parallel to the optical axis, the idea was that the weak refraction of multiple lenses might sum to this effect. However, as shown in Figure 23, the weak spherical aberration effects also sum together and the configuration is still unable to capture 100% of incoming rays at the origin.

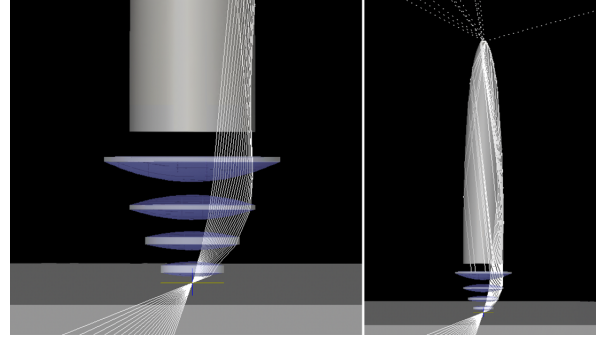


Figure 23: Stacked lenses to Winston cone

b. Sideways oriented lenses

In any vertical cross-section of BREAD, there are two outgoing angular sectors of photons each with an average angle of $(\phi_{max} + \phi_{min})/2 = 43.65^\circ = 0.762 \text{ rad}$ from the polar z-axis. In a sense, BREAD lends itself not to a focusing technique in line with the z-axis, but rather in line with this axis of average angle. A configuration that used a combination of convex lensing with a parabolic reflector was successful in capturing all the emitted ideal rays at the origin. However, these lenses would need to become the shape of a taurus to capture the full azimuthal 360° emission of photons from the cylinder and is thus not feasible to manufacture. Its performance also quickly decreases for misalignment shifts in x.

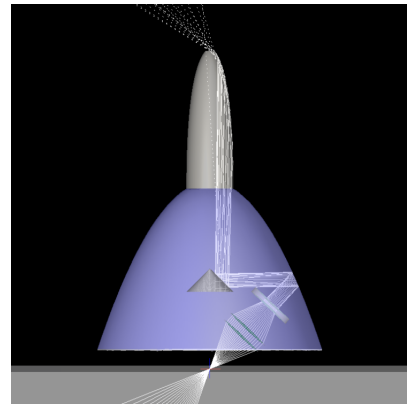


Figure 24: Sideways lens configuration into Winston cone

c. Andrew Sonnenschein's mirror-down-through-BREAD Idea

In previous talks, Dr. Andrew Sonnenschein has proposed an innovative idea of utilizing the negative space in the middle of the main BREAD reflector as a tunnel for optical focusing. At the location of the focal spot a parabolic reflector sends the rays down through the middle of the reflector either directly onto the SNSPD or onto an angled mirror that then sends the rays horizontally onto a detector. While this is a robust focusing model that works very well at the origin, any slight shifts from misalignment can drastically alter the location of the reflected rays.

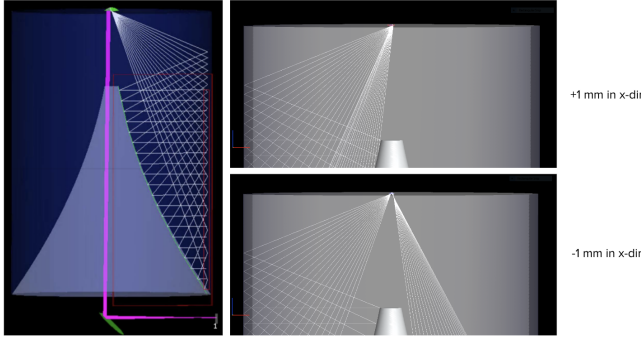


Figure 25: Mirror reflecting down middle of BREAD reflector

d. Stefan Knirck's Reverse Winston Cone to Winston Cone Idea

Dr. Stefan Knirck proposed the idea of placing an upside down Winston cone around the focal spot underneath the normal Winston cone. While this configuration also produced robust results when centered at the origin, its degree of irradiance capture quickly decreased through radial shifts. However, this was the precursor design which inspired the PTW configuration.

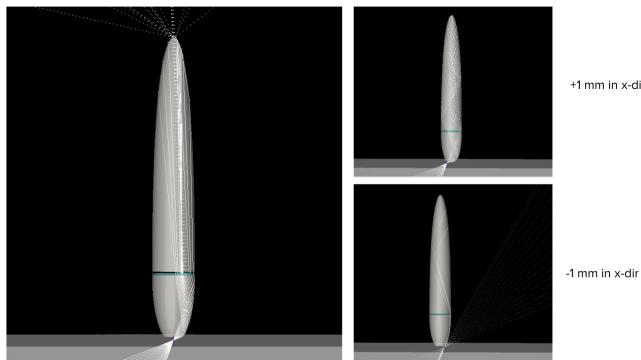


Figure 26: Reverse Winston cone to Winston cone

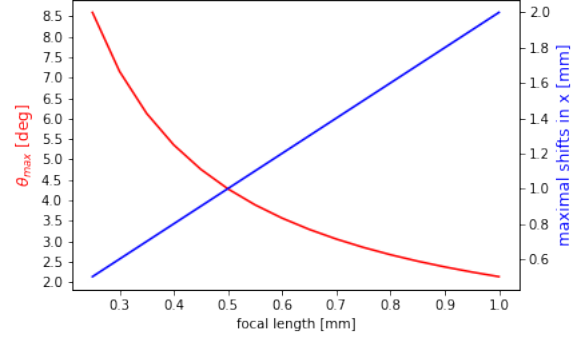


Figure 27: The inverse relationship between Winston cone angular acceptance (θ_{max}) and maximal shifts in x

Appendix B: Derivation of PTW Low and High Models

Consider a two-dimensional cross-section of BREAD along the x - z plane. In the ideal focusing scenario, the outgoing rays of BREAD will be divided into two angular sections between $[\phi_{min}, \phi_{max}]$ and $[-\phi_{min}, -\phi_{max}]$ as shown in Figure 4. Given that these outgoing rays are azimuthally symmetric around the z -axis, we can focus our attention to the section of outgoing rays with a positive r_x velocity component.

The difference between the PTW low cut and high cut models comes down to at what height, or radius, to cut the exit hole of the parabola, which will also assign the entrance hole of the Winston cone. If one cuts the exit hole of the parabola too low, then when the parabola experiences misalignment shifts in the $+x$ -direction, there will be a significant portion of outgoing rays that do not come in contact with the parabola and therefore enter the Winston cone far outside its threshold angle. However, the higher one cuts the exit hole of the parabola, the lower the angular acceptance of the Winston cone becomes. Hence, there is an optimization game to play between Winston cone threshold angle and the amount of longevity that the setup can have through shifts. Figure 27 shows this inverse relationship. Note also that each parabola can shift maximally by a distance of its entrance hole radius $r_0 = 2f$ before it overshoots the precise focal spot entirely.

Therefore, the height at which the exit hole of the parabola is cut for both the PTW low and high cut models depends on the point of intersection of the parabola and the ray path of the outgoing photon at ϕ_{min} . In the global BREAD coordinate system, this path is given by

$$z = \tan\left(\frac{\pi}{2} - \phi_{min}\right)x = \tan(1.233 \text{ rad})x \quad (\text{B1})$$

The construction of the PTW low cut model optimizes the Winston cone angular acceptance, and consequently irradiance efficiency when the optical configuration is centered at the origin (i.e. the first project goal). This is

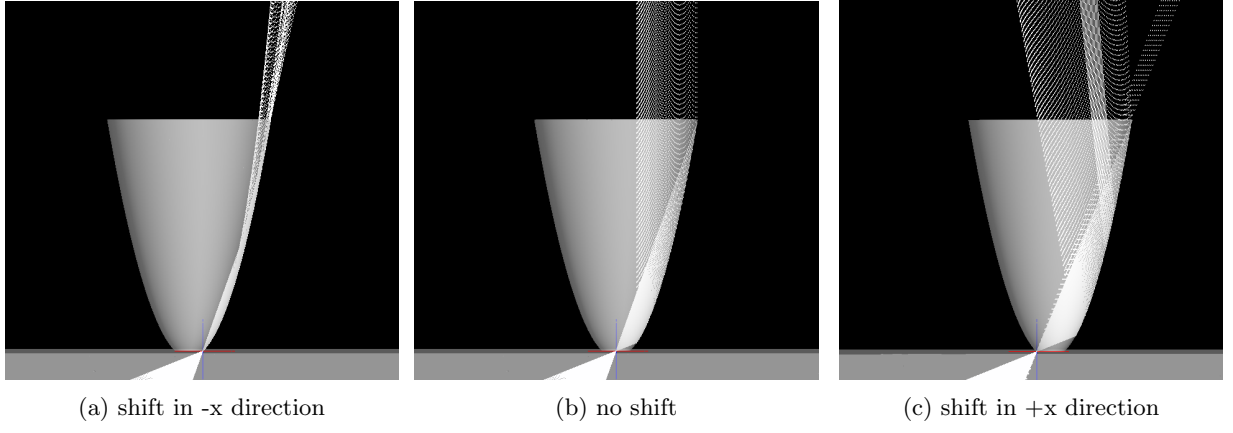


Figure 28: Low cut parabola through misalignment shifts. Figure (c) shows the fraction of outgoing photons that are not reflected by the shifted low cut parabola.

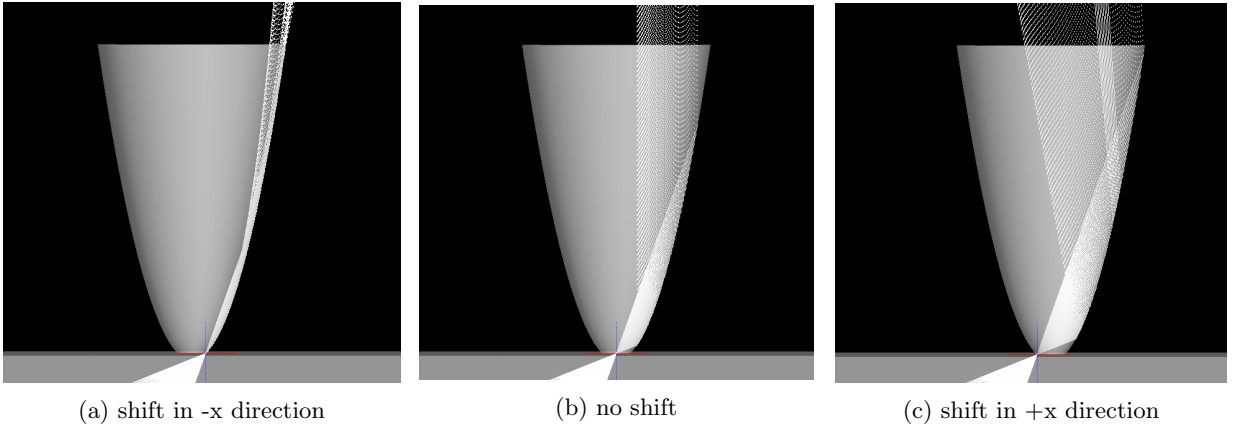


Figure 29: High cut parabola through misalignment shifts. Figure (c) shows that all of the outgoing photons are reflected by the shifted high cut parabola

achieved by minimizing the exit radius of the parabola. Hence, r_{1l} (the exit radius for the low cut model) is simply given by the x intersection point between the normal parabola for a given focal length f ,

$$l(x) = \frac{1}{4f}x^2 - f \quad (\text{B2})$$

and equation B1. This intersection occurs at $x = r_{1l} = 11.73f$, and a height of $\tan(1.233)r_{1l} = \tan(1.233)11.73f = 33.39f$. As will be made clear below, it is more useful to parametrize the exit hole cut in terms of its height on the parabola rather than the radius so we define the constant $c_l = 33.39$ and we can write the low cut height as $c_l f$.

By contrast, the construction of the PTW high cut model optimizes the longevity of the parabola through misalignment shifts (i.e. the second project goal) to its maximum displacement of $x_{max} = 2f$. Finding r_1 in this case becomes a bit more complicated. First we need to find the x intersection point between the maximally

shifted parabola,

$$h(x) = \frac{1}{4f}(x - 2f)^2 - f \quad (\text{B3})$$

and equation B1. This intersection occurs at $x = 15.74f$ and a height of $\tan(1.233)15.74f = 43.80f$. We can similarly define $c_h = 43.80$ and write this high cut height as $c_h f$. Now we need to find out at what x distance this height occurs on the the non-shifted parabola and assign this to r_{1h} :

$$\begin{aligned} \frac{1}{4f}r_{1h}^2 - f &= c_h f \\ r_{1h}^2 &= 4f^2(1 + c_h) \\ r_{1h} &= 2f\sqrt{1 + c_h} \end{aligned} \quad (\text{B4})$$

Notice that for the low cut model we can also use this parametrization based on the height constant so that

$$r_{1l} = 2f\sqrt{1 + c_l} \quad (\text{B5})$$

Now for any given focal length, we can define the entire PTW configuration in either the high cut or low

cut model just based on the constants $c_l = 33.39$ and $c_h = 43.80$, which greatly helps with simulations in FRED.

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